

Kidney Disease Diet Recommendations, Food Choices, and Foods to Avoid

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Background: CKD diets are individualized by stage and complications

There is no single "kidney diet" that fits everyone with chronic kidney disease. Diet advice changes with CKD stage, lab results, symptoms, and problems like high potassium, high phosphorus, swelling, or poor nutrition. (5 sources)

In CKD, food choices can help slow problems and reduce the buildup of substances that damaged kidneys cannot clear well, but the right diet depends on how advanced the disease is and what complications are present. As kidney function falls, patients often need to be more careful with protein, sodium, potassium, phosphorus, and sometimes fluid intake, especially in later stages or when blood tests show hyperkalemia or hyperphosphatemia. For this reason, experts recommend a personalized diet rather than one fixed eating plan for all patients with CKD. Diet plans should be tailored to disease stage, metabolic stability, comorbidities, nutritional status, and the patient's food preferences and cultural background. This is also why some patients need protein restriction, while others, such as children with CKD or undernourished patients, should avoid it. Recent guidance for moderate to advanced non-dialysis CKD supports a low-protein approach in metabolically stable patients, and also encourages eating more plant-based foods than animal-based foods, with careful monitoring in advanced CKD. Working with a clinician or kidney dietitian is therefore important, because dietary needs often change as CKD progresses. ([Boshkoska et al., 2020](#)) ([Peride et al., 2024](#)) ([Tain et al., 2024](#)) ([Kaimori et al., 2025](#)) ([Narasaki et al., 2025](#))

Evidence

(Boshkoska et al., 2020)

[Nutritional management of chronic kidney disease – a case report](#)

"A person may prevent or delay some health problems from CKD by eating recommended foods and avoiding foods high in sodium, potassium, and phosphorus. Proper information about calories, fats, proteins and fluids in allowed food products is important for a person with advanced CKD. Foods rich in proteins such as meat and dairy products break down into waste products that are removed from the blood by healthy kidneys. As CKD progresses, nutritional habits must change. A health care provider, including dietitian, may recommend a patient with reduced kidney function to choose foods very carefully."

(Peride et al., 2024)

[The Key Role of Nutritional Intervention in Delaying Disease Progression and the Therapeutic Management of Diabetic Kidney Disease—A Challenge for Physicians and Patients](#)

"Therefore, it is necessary for a personalized diet to be recommended for each patient, as in the final stages of CKD, foods with high content of potassium, phosphate, and water should be avoided."

(Tain et al., 2024)

Maternal Dietary Strategies for Improving Offspring Cardiovascular–Kidney–Metabolic Health: A Scoping Review

"Protein restriction is recommended in patients with CKD, as it retards the rate of renal function decline (Kasiske et al., 1998). Likewise, the restriction of dietary sodium is recommended for the management of CKD and its associated risks (McMahon et al., 2013). As hyperkalemia and hyperphosphatemia are common complications, restricting dietary potassium and phosphorus intake is often recommended for patients with CKD [26]. Additional calcium and vitamin D supplementation may be offered to patients with advanced CKD. Furthermore, the DASH diet, plant-based diet, and the Mediterranean diet are currently being shown to play a potential role in delaying CKD progression. All together, these studies confirm that optimized diets play a pivotal role in alleviating CKD (Chan et al., 2017)."

(Kaimori et al., 2025)

Plant-Dominant Low-Protein Diets: A Promising Dietary Strategy for Mitigating Disease Progression in People with Chronic Kidney Disease—A Comprehensive Review

"PLADO should be tailored to individual patient needs, considering factors such as the disease stage, comorbidities, dietary preferences, and cultural background (Figure 3)"

".However careful monitoring and selection of plant-based foods are needed to manage potassium and phosphorus intake, particularly in advanced CKD. If hyperkalemia occurs, new potassium-binding agents (patiromer and sodium zirconium cyclosilicate) should be considered rather than discontinuing the healthy plant-based diet (Bjune et al., 2023)"

".Some groups of patients, like children with CKD or undernourished subjects, should avoid protein restriction of any kind."

(Narasaki et al., 2025)

Should a low-protein diet and ketoanalogue supplementation be part of the management of advanced chronic kidney disease?

"Current guidelines support the role of a low-protein diet in patients with moderate to advanced CKD who are metabolically stable"

".recommend a low protein diet for patients with moderate to advanced non-dialysis dependent (NDD)-CKD who are metabolically stable to reduce risk of ESKD and death. In addition to recommending lower protein intake, the recent 2024 Kidney Disease Improving Global Outcomes CKD Guidelines include a Practice Point advising that people with CKD eat more plant-based foods than animal-based foods."

Recommended overall dietary patterns and preferred foods

For many people with CKD, the best overall pattern is a mostly plant-forward, minimally processed diet rather than a diet centered on red meat, processed foods, and salt. Good choices usually include vegetables, fruits, whole grains, legumes, fish, and unsaturated fats, with adjustments if potassium or phosphorus become a problem. (18 sources)

In general, the dietary patterns most often favored for CKD are healthy, whole-food patterns such as the Mediterranean diet, DASH-style eating, and plant-dominant diets. These patterns emphasize fruits, vegetables, legumes, whole grains, and fiber, while keeping sodium low and limiting saturated fat, refined sugar, and processed meats; they have also been linked with better cardiovascular outcomes and slower kidney decline or lower CKD risk in many studies. ([Cases et al., 2019](#)) ([Kim et al., 2020](#)) ([Bach et al., 2019](#)) ([Molina et al., 2021](#)) ([Kelly et al., 2020](#)) ([Rysz et al., 2017](#)) ([Nogueira-Rio et al., 2025](#))

A practical way to eat is to build meals mostly from minimally processed foods and home-cooked dishes: vegetables, fruits, rice or other grains, beans or other legumes when allowed, and modest portions of lean protein. When available and tolerated, whole grains, high-fiber carbohydrates, and mixed fruits and vegetables are preferred over refined starches and sugary foods, and fats should come mainly from unsaturated sources such as olive, canola, or flaxseed oils rather than butter or other saturated fats. ([Mafra et al., 2020](#)) ([Kalantar-Zadeh et al., 2017](#)) ([Kang et al., 2025](#)) ([Nogueira-Rio et al., 2025](#))

Many recent reviews and guidelines also favor shifting more of the diet toward plant foods than animal foods, especially replacing red and processed meat with vegetables, grains, legumes, and other plant foods. This can lower dietary acid load and may put less stress on the kidneys; fruit and vegetable intake may also help counter metabolic acidosis, which can worsen with high intakes of meat and refined cereals. ([Kim et al., 2020](#)) ([Kim et al., 2019](#)) ([Cases et al., 2019](#)) ([Moldovan et al., 2024](#)) ([Pradhan et al., 2025](#)) ([Raju et al., 2025](#))

Preferred protein-containing foods within this overall pattern are usually fish, poultry, and plant protein sources, with attention to the patient's full nutrition prescription. Fish is often encouraged in place of red and processed meat, and several sources describe plant-dominant low-protein patterns that use at least half of protein from plant sources and rely on unprocessed foods. ([Kochan et al., 2021](#)) ([Guldris et al., 2022](#)) ([Mafra et al., 2020](#)) ([Hu et al., 2025](#))

So, foods that are often good staples for CKD patients include fresh or low-sodium vegetables and fruits, whole grains or other high-fiber starches, beans and legumes if they fit potassium and phosphorus goals, fish, small amounts of lean meats or poultry, and unsaturated fats. If fresh produce is hard to get, low-sodium and low-sugar canned fruits and vegetables can also be used. ([Mafra et al., 2020](#)) ([Guldris et al., 2022](#)) ([Hu et al., 2025](#)) ([Liu et al., 2025](#))

At the same time, the overall pattern should move away from ultra-processed foods and additives, especially foods high in sodium, refined sugar, and processed meat. This means eating sausages, cold cuts, fast food, packaged snacks, sugar-sweetened drinks, cakes, cookies, and similar foods only rarely, if at all, because these foods tend to worsen diet quality and can add sodium, highly absorbable phosphorus additives, and excess sugar. ([Mafra et al., 2020](#)) ([Moldovan et al., 2024](#)) ([Pradhan et al., 2025](#)) ([Raju et al., 2025](#))

One important caution is that even healthy patterns like DASH, Mediterranean, or plant-dominant diets may be high in potassium, and sometimes phosphorus, so they need to be adapted to the person's lab results rather than copied exactly. That is why a "healthy CKD diet" usually means a balanced, mostly plant-forward pattern with careful food selection, not simply eating unlimited fruits, vegetables, nuts, legumes, or dairy. ([Guldris et al., 2022](#)) ([Kim et al., 2020](#)) ([Mafra et al., 2020](#))

Evidence

(Cases et al., 2019)

[Vegetable-Based Diets for Chronic Kidney Disease? It Is Time to Reconsider](#)

"Many studies recommend plant-based diets, such as the Dietary Approaches to Stop Hypertension (DASH) diet or the Mediterranean diet (MD), which are rich in fiber; are low in saturated fat and processed meats; contain sources of potassium, phosphorus, magnesium, and calcium; and have a low sodium content. MD and DASH diets are similar, both have been associated with a lower risk of mortality from cardiovascular disease and slow the progression of kidney disease (Chauveau et al., 2018)"

"Metabolic acidosis in chronic kidney disease (CKD) is aggravated by the high consumption of meat and refined cereals, increasing the dietary acid load, while the intake of fruit and vegetables is able to neutralize the acidosis and its deleterious consequences."

(Kim et al., 2020)

[Nutritional management in patients with chronic kidney disease](#)

"Many guidelines recommend limiting protein intake in these patients, as high-protein diets aggravate kidney dysfunction. Excess sodium may be associated with CKD progression and all-cause mortality and, therefore, limiting salt intake is generally recommended"

"Previous observational studies have shown that a plant-based diet is associated with a slower decline in the eGFR cr among patients with CKD (Bach et al., 2019)(Kim et al., 2019). Eating more plant-based foods, such as vegetables and grains, in place of animal-based foods, such as red meat, may help prevent and slow the progression of CKD, type 2 diabetes, high blood pressure, and heart disease (Bach et al., 2019)(Kim et al., 2019). Eating less animal-based foods can lower the acid load and put less stress on the kidneys (Bach et al., 2019)(Kim et al., 2019)"

"Most recent guidelines recommend adjusting dietary potassium intake on an individual basis to maintain normal serum potassium levels in patients with CKD [40]."

(Bach et al., 2019)

[Healthy Dietary Patterns and Incidence of CKD: A Meta-Analysis of Cohort Studies.](#)

"BACKGROUND AND OBJECTIVES

Whether a healthy dietary pattern may prevent the incidence of developing CKD is unknown. This study evaluated the associations between dietary patterns and the incidence of CKD in adults and children.

DESIGN, SETTING, PARTICIPANTS, & MEASUREMENTS

This systematic review and meta-analysis identified potential studies through a systematic search of MEDLINE, Embase and references from eligible studies from database inception to February 2019. Eligible studies were prospective and retrospective cohort studies including adults and children without CKD, where the primary exposure was dietary patterns. To be eligible, studies had to report on the primary outcome, incidence of CKD (eGFR<60 ml/min per 1.73 m²). Two authors independently extracted data, assessed risk of bias and evidence certainty using the Newcastle-Ottawa scale and GRADE.

RESULTS

Eighteen prospective cohort studies involving 630,108 adults (no children) with a mean follow-up of 10.4±7.4 years were eligible for analysis. Included studies had an overall low risk of bias. The evidence certainty was moderate for CKD incidence and low for eGFR decline (percentage drop from baseline or reduced by at least 3 ml/min per 1.73 m² per year) and incident albuminuria. Healthy dietary patterns typically encouraged higher intakes of vegetables, fruit, legumes, nuts, whole grains, fish and low-fat dairy, and lower intakes of red and processed meats, sodium, and sugar-sweetened beverages. A healthy dietary pattern was associated with a lower incidence of CKD (odds ratio [OR] 0.70 [95% confidence interval [95% CI], 0.60 to 0.82]; I²=51%; eight studies), and incidence of albuminuria (OR 0.77, [95% CI, 0.59 to 0.99]; I²=37%; four studies). There appeared to be no significant association between healthy dietary patterns and eGFR decline (OR 0.70 [95% CI, 0.49 to 1.01], I²=49%; four studies).

CONCLUSIONS

A healthy dietary pattern may prevent CKD and albuminuria."

(Molina et al., 2021)

Optimizing Diet to Slow CKD Progression

"Such patients should be advised to follow the same dietary recommendations as for the general population [low sodium and refined sugar, avoidance of red and processed meats, and high content of fruits, vegetables, legumes, fish, poultry, and whole grains (Kelly et al., 2020)]."

(Kelly et al., 2020)

Modifiable Lifestyle Factors for Primary Prevention of CKD: A Systematic Review and Meta-Analysis.

"BACKGROUND

Despite increasing incidence of CKD, no evidence-based lifestyle recommendations for CKD primary prevention apparently exist.

METHODS

To evaluate the consistency of evidence associating modifiable lifestyle factors and CKD incidence, we searched MEDLINE, Embase, CINAHL, and references from eligible studies from database inception through June 2019. We included cohort studies of adults without CKD at baseline that reported lifestyle exposures (diet, physical activity, alcohol consumption, and tobacco smoking). The primary outcome was incident CKD (eGFR<60 ml/min per 1.73 m²). Secondary outcomes included other CKD surrogate measures (RRT, GFR decline, and albuminuria).

RESULTS

We identified 104 studies of 2,755,719 participants with generally a low risk of bias. Higher dietary potassium intake associated with significantly decreased odds of CKD (odds ratio [OR], 0.78; 95% confidence interval [95% CI], 0.65 to 0.94), as did higher vegetable intake (OR, 0.79; 95% CI, 0.70 to 0.90); higher salt intake associated with significantly increased odds of CKD (OR, 1.21; 95% CI, 1.06 to 1.38). Being physically active versus sedentary associated with lower odds of CKD (OR, 0.82; 95% CI, 0.69 to 0.98). Current and former smokers had significantly increased odds of CKD compared with never smokers (OR, 1.18; 95% CI, 1.10 to 1.27). Compared with no consumption, moderate consumption of alcohol associated with reduced risk of CKD (relative risk, 0.86; 95% CI, 0.79 to 0.93). These associations were consistent, but evidence was predominantly of low to very low certainty. Results for secondary outcomes were consistent with the primary finding.

CONCLUSIONS

These findings identify modifiable lifestyle factors that consistently predict the incidence of CKD in the community and may inform both public health recommendations and clinical practice."

(Rysz et al., 2017)

The Effect of Diet on the Survival of Patients with Chronic Kidney Disease

"A recent systemic study revealed that a healthy diet comprising many fruits and vegetables, fish, legumes, whole grains, and fibers

and also the cutting down on red meat, sodium, and refined sugar intake was associated with lower mortality in people with kidney disease."

(Nogueira-Rio et al., 2025)

Rethinking Nutrition in Chronic Kidney Disease: Plant Foods, Bioactive Compounds, and the Shift Beyond Traditional Limitations: A Narrative Review

"The regular consumption of plant-based foods such as cereals, raw vegetables, cooked vegetables, and fresh fruit is among the main nutritional factors associated with the prevention of CKD [21]. Conversely, high consumption of processed meat, white bread, and other less healthy foods is associated with an increased risk of CKD episodes [21]"

"This energy should come mainly from healthy macronutrients, so simple carbohydrates (sugars) should be avoided, and complex carbohydrates rich in fiber should be consumed. For lipids, the consumption of monounsaturated and polyunsaturated fatty acids is recommended instead of trans fats or saturated fatty acids, which increase the risk of CVD [23]. Among dietary patterns, preliminary results from prospective studies have shown that better adherence to a Mediterranean or dietary approach to stop hypertension (DASH) type diet is associated with a lower incidence of severe kidney damage and a lower risk of kidney function decline (Herber-Gast et al., 2017)."

(Mafra et al., 2020)

Coronavirus Disease 2019: Quick Diet and Nutrition Guide for Patients With Chronic Kidney Disease

"Patients with CKD should consume minimally processed natural or whole foods such as fruits, vegetables, rice, lean meats, and home-cooked meals (instead of buying foods from a restaurant or processed foods from the market). However, when fresh fruits and vegetables are not available, canned vegetables and fruits low in sodium and sugar can be included in the diet. On the other hand, most of the processed foods with excess amounts of sodium, such as sausages and cold cuts, and excess sugar, such as cakes, cookies, candy, and sugar-sweetened beverages, should be consumed rarely. Milk and other dairy products, legumes (e.g., beans), and high-fat meats should be limited in the diet according to the diet prescription. (Mafra et al., 2016)"

"Intake of foods high in potassium (K⁺) should be monitored in patients with high serum potassium levels. (Ikizler et al., 2020)"

"Patients who live in tropical areas are prohibited from eating carambolas, also known as star fruit, given the accumulation of toxic substances. 16 It is important to remember that a low-protein diet (0.55-0.60 g dietary protein/kg ideal body weight/day) 12 and a plant-dominant low-protein diet with more than 50% plant-based sources are recommended for nondialysis patients with CKD"

(Kalantar-Zadeh et al., 2017)

Nutritional Management of Chronic Kidney Disease

"In patients with kidney disease, carbohydrates should be complexed with high fiber content (e.g., whole-wheat breads, multigrain cereal, oatmeal, and mixed fruits and vegetables) to help reduce dietary phosphorus and protein as well as urea and creatinine generation. (Chiavaroli et al., 2014)"

"Nonsaturated fat is the preferred lipid in the diet. Replacement of butter with flaxseed, canola, or olive oil, all of which are rich in n-3 fatty acids, may be worthwhile. (Khoueiry et al., 2011)"

(Kang et al., 2025)

Nutritional Status Evaluation and Intervention in Chronic Kidney Disease Patients: Practical Approach

"Energy provision should prioritize nutrient-dense sources with emphasis on healthy fats (mono-and polyunsaturated fatty acids), complex carbohydrates, and fiber, while minimizing simple sugars and ultra-processed foods to reduce metabolic stress"

". In patients with CKD, low protein diets (LPDs) are frequently prescribed as a therapeutic strategy to reduce nitrogenous waste products, control uremic symptoms, and slow the progression of renal function decline."

(Kim et al., 2019)

Plant-Based Diets and Incident CKD and Kidney Function.

"BACKGROUND AND OBJECTIVES

The association between plant-based diets, incident CKD, and kidney function decline has not been examined in the general population. We prospectively investigated this relationship in a population-based study, and evaluated if risk varied by different types of plant-based diets.

DESIGN, SETTING, PARTICIPANTS, & MEASUREMENTS

Analyses were conducted in a sample of 14,686 middle-aged adults enrolled in the Atherosclerosis Risk in Communities study. Diets were characterized using four plant-based diet indices. In the overall plant-based diet index, all plant foods were positively scored; in

the healthy plant-based diet index, only healthful plant foods were positively scored; in the provegetarian diet, selected plant foods were positively scored. In the less healthy plant-based diet index, only less healthful plant foods were positively scored. All indices negatively scored animal foods. We used Cox proportional hazards models to study the association with incident CKD and linear mixed models to examine decline in eGFR, adjusting for confounders.

RESULTS

During a median follow-up of 24 years, 4343 incident CKD cases occurred. Higher adherence to a healthy plant-based diet (HR comparing quintile 5 versus quintile 1 [HRQ5 versus Q1], 0.86; 95% confidence interval [95% CI], 0.78 to 0.96; P for trend =0.001) and a provegetarian diet (HRQ5 versus Q1, 0.90; 95% CI, 0.82 to 0.99; P for trend =0.03) were associated with a lower risk of CKD, whereas higher adherence to a less healthy plant-based diet (HRQ5 versus Q1, 1.11; 95% CI, 1.01 to 1.21; P for trend =0.04) was associated with an elevated risk. Higher adherence to an overall plant-based diet and a healthy plant-based diet was associated with slower eGFR decline. The proportion of CKD attributable to lower adherence to healthy plant-based diets was 4.1% (95% CI, 0.6% to 8.3%).

CONCLUSIONS

Higher adherence to healthy plant-based diets and a vegetarian diet was associated with favorable kidney disease outcomes."

(Moldovan et al., 2024)

Food to Prevent Vascular Calcification in Chronic Kidney Disease

"The foundation of nutrition intervention in CKD was laid for decades on a low-protein diet to slow progression and on restriction of plant foods, such as vegetables and fruits, to prevent hyperkalemia. Lately, this paradigm has changed, and the plant-dominant low-protein (PLADO) diets seem to have become a better choice for patients with CKD (Chen et al., 2022)"

".Such a diet is based on fruits, vegetables, seeds, nuts, tea, cocoa, and whole-grain cereals (Carrero et al., 2020)"

".Evidence reports the harmfulness of ultra-processed food, food additives, and animal-based proteins due to the increased intake of high absorbable phosphorus, the scarcity of fibers, and the increased production of uremic toxins."

(Pradhan et al., 2025)

Personalized Nutrition in Chronic Kidney Disease

"The Kidney Disease: Improving Global Outcomes (KDIGO) 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease recommends that patients with CKD adopt healthy and diverse diets with greater consumption of plant-based foods compared to animal-based foods and reduce the intake of ultraprocessed foods [21]"

".The American Diabetes Association (ADA) and KDIGO consensus report on diabetes management in CKD emphasizes the importance of individualized and balanced diets that are high in vegetables, fruits, and whole grains but low in refined carbohydrates and sugar-sweetened beverages. Both guidelines recommend a low-sodium diet (KDIGO < 2000 mg/day; ADA 1500 to <2300 mg/day) to control blood pressure and reduce cardiovascular risk [23]."

(Raju et al., 2025)

Hyperphosphatemia in Kidney Failure: Pathophysiology, Challenges, and Critical Role of Phosphorus Management

"The International Society of Renal Nutrition and Metabolism (ISRNM) recommends incorporation of plant-based protein sources into the dietary plans of CKD/KF patients while considering individual preferences and cultural dietary habits (Kistler et al., 2020). Likewise, the 2024 KDIGO Clinical Practice Guideline for the Evaluation and Management of CKD emphasizes a balanced, diverse diet, prioritizing plant-based foods while limiting animal proteins and processed foods (Stevens et al., 2024). Hence, a well-planned diet should minimize processed foods and additives and emphasize plant-based sources."

(Kochan et al., 2021)

Dietary Lipids and Dyslipidemia in Chronic Kidney Disease

"The most important elements of the diet are an adequate energy intake of about 25-35 kcal/kg/d, adjusted to maintain normal nutritional status; protein content in the diet, which should decrease to 0.55-0.6 g/kg/d with the loss of kidney function but should increase to 1.0-1.2 g/kg/d once dialysis is initiated; and the restriction of dietary sodium, phosphate, and potassium (Ikizler et al., 2020)"

".Patients with CKD should therefore be encouraged to increase fish consumption. Nutritional education of CKD patients by professionals is advised to individualize the diet and to improve the patients' food choices to favor unsaturated fats, especially long-chain ω -3 PUFAs from carefully chosen species of oily fish, instead of saturated fats from red and processed meat. Patients with advanced CKD are at substantial risk for malnutrition, so they most likely benefit from well-balanced healthy dietary patterns such as the Mediterranean diet, DASH, or plant-based diets"

(Guldris et al., 2022)

Fibre Intake in Chronic Kidney Disease: What Fibre Should We Recommend?

"Healthy dietary patterns (a higher intake of fruits, vegetables, legumes, and fish and a reduction in the intake of red meat, sodium and refined sugar) are associated with lower mortality in people with CKD"

". Plant dominant low protein diet (PLADO). It is a low-protein diet (0.6-0.8 g/kg/day) with at least 50% plant-based protein, based on unprocessed foods, a low sodium intake of <3 g/day, fibre from 25 to 30 g/day and caloric intake of 30-35 Kcal/kg/day"

". These three dietary patterns have high potassium quantities, which should be adjusted to maintain potassium levels within the normal range, as indicated by the KDOQI 2020 guide."

(Hu et al., 2025)

Individualized Diets in Patients with Kidney Disease and Kidney Transplants: A Narrative Review

"A balanced Mediterranean diet, rich in fruits, vegetables, whole grains, fish, and unsaturated fats, shows promises in improving the prognosis for CKD patients. Plant-based diets, which emphasize legumes, vegetables, and grains while minimizing animal protein, improve blood pressure and the glycemic and lipid control. Low-protein diets (LPDs), typically providing less than 0.6 g/kg/day of protein, may reduce the CKD progression and nitrogen burden, further delaying the initiation of dialysis."

(Liu et al., 2025)

Uric Acid and Atherosclerosis in Patients with Chronic Kidney Disease: Recent Progress, Mechanisms, and Prospect

"A balanced, healthy diet is important to control serum UA, especially for patients with kidney disease (Choi et al., 2004). Fresh vegetables, eggs, low-fat, or skim milk are recommended. In the meantime, food rich in purines should be limited"

".Patients are allowed to consume fruits with low fructose contents, such as strawberries, pineapples, watermelons, and peaches"

".Alcohol should be limited for the effect of increasing risk of hyperuricemia, and beer, yellow rice wine, and spirits should be forbidden (Choi et al., 2004)(Wang et al., 2013). Smoking and passive smoking should both be avoided."

Protein intake and protein sources

In non-dialysis CKD, protein often needs to be moderated rather than increased. Many guidelines support about 0.8 g/kg/day for metabolically stable adults with stages 3 to 5 CKD, avoiding high-protein diets and often favoring more plant protein over red and processed meat. (14 sources)

Protein is one of the most important parts of a CKD diet because both too much and too little can cause problems. In metabolically stable adults with CKD stages 3 to 5 who are not on dialysis, guidelines support protein restriction, commonly around 0.8 g/kg/day, to help delay kidney failure and improve symptoms or quality of life. Some reviews describe low-protein diets in the range of about 0.6 to 0.8 g/kg/day, and very low-protein diets around 0.3 to 0.4 g/kg/day are sometimes used in selected predialysis patients, usually with ketoacid or essential amino acid support and close supervision. ([Cupisti et al., 2020](#)) ([Ko et al., 2017](#)) ([Sakaguchi et al., 2023](#)) ([Dabek et al., 2023](#)) ([Martinez-Hernandez et al., 2023](#)) ([Hahn et al., 2018](#)) ([Rodelo-Haad et al., 2025](#))

Just as important, CKD patients generally should avoid high-protein eating patterns, especially diets that replace carbohydrates with large amounts of meat or other protein foods. High protein intake, particularly above about 1.3 g/kg/day in adults at risk of CKD progression, has been linked to intraglomerular hypertension, hyperfiltration, proteinuria, and faster kidney damage, so high-protein weight-loss diets are usually a poor fit for CKD. ([Sakaguchi et al., 2023](#)) ([Wang et al., 2024](#)) ([Ko et al., 2020](#)) ([Ko et al., 2017](#))

When choosing protein foods, many sources favor shifting away from animal-heavy patterns and toward more plant-based protein. Plant-dominant low-protein approaches, often with at least half of protein coming from plant sources, may help by lowering acid load, reducing gut-derived uremic toxins, improving phosphorus control, and causing less glomerular hyperfiltration than animal protein. In practical terms, this usually means using more beans, lentils, soy foods, and other plant proteins when allowed, and choosing fish or poultry more often than red or processed meat. ([Sakaguchi et al., 2023](#)) ([Zhu et al., 2025](#)) ([Kalantar-Zadeh et al., 2020](#)) ([Wang et al., 2024](#)) ([Orczykowski et al., 2022](#))

That said, protein restriction is not right for every patient. If someone is undernourished, losing weight unintentionally, has protein-energy wasting, or has other signs that nutrition is poor, preserving protein intake may be more important than restricting it. Older adults need especially careful balancing, and some patients should avoid or postpone protein restriction until nutritional status is stable. ([Dabek et al., 2023](#)) ([Piccoli et al., 2023](#)) ([Rodelo-Haad et al., 2025](#))

So, for most non-dialysis CKD patients, the practical message is: do not follow a high-protein diet; aim for a clinician-guided moderate or low protein target; get more protein from plant foods, fish, and poultry; and keep red and processed meats smaller and less frequent. Because protein targets change with CKD stage, dialysis status, and nutrition risk, the exact amount should be set with a kidney dietitian or clinician rather than guessed. ([Cupisti et al., 2020](#)) ([Sakaguchi et al., 2023](#)) ([George et al., 2018](#))

Evidence

(Cupisti et al., 2020)

[Medical Nutritional Therapy for Patients with Chronic Kidney Disease not on Dialysis: The Low Protein Diet as a Medication](#)

"The 2020 Kidney Disease Outcome Quality Initiative (KDOQI) Clinical Practice Guideline for Nutrition in chronic kidney disease (CKD) recommends protein restriction to patients affected by CKD in stages 3 to 5 (not on dialysis), provided that they are metabolically stable, with the goal to delay kidney failure (graded as evidence level 1A) and improve quality of life (graded as evidence level 2C)."

(Ko et al., 2017)

[Dietary Approaches in the Management of Diabetic Patients with Kidney Disease](#)

"expert panels recommend lower dietary protein intake of 0.8 g/kg of body weight/day"

"Carbohydrates from sugars should be limited to less than 10% of energy intake, and it is also suggested that higher polyunsaturated and monounsaturated fat consumption in lieu of saturated fatty acids, trans-fat, and cholesterol are associated with more favorable outcomes. While guidelines recommend dietary sodium restriction to less than 1.5–2.3 g/day"

"However, a low carbohydrate diet that is replaced by high dietary protein sources should be avoided in DKD patients (Otoda et al., 2014)"

(Sakaguchi et al., 2023)

[Plant-Dominant Low Protein Diet: A Potential Alternative Dietary Practice for Patients with Chronic Kidney Disease](#)

"Therefore, protein restriction has long been a cornerstone of nutritional therapy for patients with CKD"

".this evidence may support the Kidney Disease: Improving Global Outcomes (KDIGO) 2012 CKD guideline that recommends to avoid a protein intake of >1.3 g/kg per day in adults with CKD at risk of progression in order to prevent the deterioration of kidney function [7]"

".Furthermore, a plant-dominant low-protein diet (PLADO) has recently been proposed as an alternative dietary therapy for patients with CKD. There are several potential mechanisms by which plant-based diets would benefit patients with CKD. For example, plant-based diets may reduce the production of gut-derived uremic toxins by increasing the intake of fiber, and are useful for correcting metabolic acidosis and hyperphosphatemia. Plant proteins are less likely to induce glomerular hyperfiltration than animal proteins."

(Dabek et al., 2023)

[Novel Therapeutic Approaches in the Management of Chronic Kidney Disease](#)

"The recommended dietary allowances for individuals with CKD advise adhering to a low protein intake (ranging from 0.3 to 0.8 g per kilogram of body weight per day), restricting sodium to less than 2.3 g per day, limiting phosphate intake to below 700 milligrams per day, and considering supplementation of essential amino acids, ketoacids, calcium carbonate, vitamins, and iron (Piccoli et al., 2023)."

(Martinez-Hernandez et al., 2023)

[Novel Approaches in Chronic Renal Failure without Renal Replacement Therapy: A Review](#)

"In most instances, very low-protein diets (VLPD) are recommended. Therefore, supplementation with ketoanalogues, such as ketoskeril, becomes essential for predialysis patients (Garneata et al., 2016)"

"In CKD, diets featuring a predominance of vegetables (e.g., the Mediterranean diet) are recommended"

"Significant reductions in protein, phosphorus, calcium, and potassium intake can render diets less wholesome, and in some cases, more detrimental than beneficial (Hahn et al., 2018)(Kiuchi et al., 2016)."

(Hahn et al., 2018)

Low protein diets for non-diabetic adults with chronic kidney disease.

"BACKGROUND

Chronic kidney disease (CKD) is defined as reduced function of the kidneys present for 3 months or longer with adverse implications for health and survival. For several decades low protein diets have been proposed for participants with CKD with the aim of slowing the progression to end-stage kidney disease (ESKD) and delaying the onset of renal replacement therapy. However the relative benefits and harms of dietary protein restriction for preventing progression of CKD have not been resolved. This is an update of a systematic review first published in 2000 and updated in 2006 and 2009.

OBJECTIVES

To determine the efficacy of low protein diets in preventing the natural progression of CKD towards ESKD and in delaying the need for commencing dialysis treatment in non-diabetic adults.

SEARCH METHODS

We searched the Cochrane Kidney and Transplant Register of Studies up to 2 March 2018 through contact with the Information Specialist using search terms relevant to this review. Studies in the Register are identified through searches of CENTRAL, MEDLINE, and EMBASE, conference proceedings, the International Clinical Trials Register (ICTRP) Search Portal and ClinicalTrials.gov.

SELECTION CRITERIA

We included randomised controlled trials (RCTs) or quasi RCTs in which adults with non-diabetic chronic kidney disease (stages 3 to 5) not on dialysis were randomised to receive a very low protein intake (0.3 to 0.4 g/kg/d) compared with a low protein intake (0.5 to 0.6 g/kg/d) or a low protein intake compared with a normal protein intake (≥ 0.8 g/kg/d) for 12 months or more.

DATA COLLECTION AND ANALYSIS

Two authors independently selected studies and extracted data. For dichotomous outcomes (death, all causes), requirement for dialysis, adverse effects) the risk ratios (RR) with 95% confidence intervals (CI) were calculated and summary statistics estimated using the random effects model. Where continuous scales of measurement were used (glomerular filtration rate (GFR), weight), these data were analysed as the mean difference (MD) or standardised mean difference (SMD) if different scales had been used. The certainty of the evidence was assessed using GRADE.

MAIN RESULTS

We identified an additional six studies to include 17 studies with 2996 analysed participants (range 19 to 840). Four larger multicentre studies were subdivided according to interventions so that the review included 21 separate data sets. Mean duration of participant follow-up ranged from 12 to 50 months. Random sequence generation and allocation concealment were considered at low risk of bias in eleven and nine studies respectively. All studies were considered at high risk for performance bias as they were open-label studies. We assessed detection bias for outcome assessment for GFR and ESKD separately. As GFR measurement was a laboratory outcome all studies were assessed at low risk of detection bias. For ESKD, nine studies were at low risk of detection bias as the need to commence dialysis was determined by personnel independent of the study investigators. Five studies were assessed at high risk of attrition bias with eleven studies at low risk. Ten studies were at high risk for reporting bias as they did not include data which could be included in a meta-analysis. Eight studies reported funding from government bodies while the remainder did not report on funding. Ten studies compared a low protein diet with a normal protein diet in participants with CKD categories 3a and b (9 studies) or 4 (one study). There was probably little or no difference in the numbers of participants who died (5 studies 1680 participants: RR 0.77, 95% CI 0.51 to 1.18; 13 fewer deaths per 1000; moderate certainty evidence). A low protein diet may make little or no difference in the number of participants who reached ESKD compared with a normal protein diet (6 studies, 1814 participants: RR 1.05, 95% CI 0.73 to 1.53; 7 more per 1000 reached ESKD; low certainty evidence). It remains uncertain whether a low protein diet compared with a normal protein intake impacts on the outcome of final or change in GFR (8 studies, 1680 participants: SMD -0.18, 95% CI -0.75 to 0.38; very low certainty evidence). Eight studies compared a very low protein diet with a low protein diet and two studies compared a very low protein diet with a normal protein diet. A very low protein intake compared with a low protein intake probably made little or no difference to death (6 studies, 681 participants: RR 1.26, 95% CI 0.62 to 2.54; 10 more deaths per 1000; moderate certainty evidence). However it probably reduces the number who reach ESKD (10 studies, 1010 participants: RR 0.65, 95% CI 0.49 to 0.85; 165 per 1000 fewer reached ESKD; moderate certainty evidence). It remains uncertain whether a very low protein diet compared with a low or normal protein intake influences the final or change in GFR (6 studies, 456 participants: SMD 0.12, 95% CI -0.27 to 0.52; very low certainty evidence). Final body weight was reported in only three studies. It is

uncertain whether the intervention alters final body weight (3 studies, 89 participants: MD -0.40 kg, 95% CI -6.33 to 5.52; very low certainty evidence). Twelve studies reported no evidence of protein energy wasting (malnutrition) in their study participants while three studies reported small numbers of participants in each group with protein energy wasting. Most studies reported that adherence to diet was satisfactory. Quality of life was not formally assessed in any studies.

AUTHORS' CONCLUSIONS

This review found that very low protein diets probably reduce the number of people with CKD 4 or 5, who progress to ESKD. In contrast low protein diets may make little difference to the number of people who progress to ESKD. Low or very low protein diets probably do not influence death. However there are limited data on adverse effects such as weight differences and protein energy wasting. There are no data on whether quality of life is impacted by difficulties in adhering to protein restriction. Studies evaluating the adverse effects and the impact on quality of life of dietary protein restriction are required before these dietary approaches can be recommended for widespread use."

(Rodelo-Haad et al., 2025)

[The true cost of phosphate control in chronic kidney disease](#)

"The latest KDIGO guidelines recommend a reduction in the consumption of ultra-processed foods and suggest that adults with CKD stages 3-5 maintain a protein intake of 0.8 g protein per kg body weight per day [125]"

"It is preferable to reduce the intake of inorganic phosphate rather than limiting the intake of phosphate contained in proteins, which are necessary to prevent poor nutrition and muscle wasting. Therefore, protein restriction should not be recommended in patients with an elevated P/UUN ratio; rather, in these patients, protein intake should be preserved, and the easily absorbable inorganic phosphate should be restricted. It is essential to make individualized dietary recommendations to reduce the risk of phosphate-related renal progression while maintaining good nutritional status."

(Wang et al., 2024)

[Recent Advances in the Management of Diabetic Kidney Disease: Slowing Progression](#)

"High dietary protein intake can lead to intraglomerular hypertension, which contributes to glomerular hyperfiltration, kidney damage and proteinuria (Ko et al., 2020)"

". Therefore, patients with advanced CKD are advised to restrict their potassium intake. Lower-potassium fruits, vegetables, and other foods are recommended for DKD patients, and the intake of vegetables and fruits should be in accordance with normal diabetic diet recommendations"

". Therefore, healthy diets, including fruits, vegetables, whole grains, legumes, fiber, unsaturated fats, plant-based proteins, and nuts have been revealed to be associated with lower incidences of CKD and albuminuria [180]. Additionally, restricting sodium intake is associated with a significant decrease in the risk of stroke, cardiovascular disease, and progression of CKD [181]."

(Ko et al., 2020)

[The Effects of High-Protein Diets on Kidney Health and Longevity.](#)

"Although high-protein diets continue to be popular for weight loss and type 2 diabetes, evidence suggests that worsening renal function may occur in individuals with-and perhaps without-impaired kidney function. High dietary protein intake can cause intraglomerular hypertension, which may result in kidney hyperfiltration, glomerular injury, and proteinuria. It is possible that long-term high protein intake may lead to de novo CKD. The quality of dietary protein may also play a role in kidney health. Compared with protein from plant sources, animal protein has been associated with an increased risk of ESKD in several observational studies, including the Singapore Chinese Health Study. Potential mediators of kidney damage from animal protein include dietary acid load, phosphate content, gut microbiome dysbiosis, and resultant inflammation. In light of such findings, adopting current dietary approaches that include a high proportion of protein for weight reduction or glycemic control should be considered with care in those at high risk for kidney disease. Given the possibility of residual confounding within some observational studies and the conflicting evidence from previous trials, long-term studies including those with large sample sizes are warranted to better ascertain the effects of high protein intake on kidney health."

(Zhu et al., 2025)

[The influence of gut microbiota on the gut-brain-kidney axis and its implications for chronic kidney disease](#)

"Limiting the intake of sodium, protein, phosphorus, and potassium is generally recommended in patients with CKD, while foods rich in fiber, vitamins, and minerals are greatly advised (Yang et al., 2018)(Mafra et al., 2020). As a result, adequate care should be taken in exploring high-fiber foods, as low-potassium-containing high-fiber foods for CKD patients are the goal. By lowering phosphorus load, metabolic acidosis, and uremic toxin production, a low-protein diet (LPD) may improve CKD-associated symptoms and slow renal function deterioration (Kalantar-Zadeh et al., 2020)Kalantar-Zadeh et al., 2022)."

(Kalantar-Zadeh et al., 2020)

Plant-Dominant Low-Protein Diet for Conservative Management of Chronic Kidney Disease

"Chronic kidney disease (CKD) affects >10% of the adult population. Each year, approximately 120,000 Americans develop end-stage kidney disease and initiate dialysis, which is costly and associated with functional impairments, worse health-related quality of life, and high early-mortality rates, exceeding 20% in the first year. Recent declarations by the World Kidney Day and the U.S. Government Executive Order seek to implement strategies that reduce the burden of kidney failure by slowing CKD progression and controlling uremia without dialysis. Pragmatic dietary interventions may have a role in improving CKD outcomes and preventing or delaying dialysis initiation. Evidence suggests that a patient-centered plant-dominant low-protein diet (PLADO) of 0.6–0.8 g/kg/day composed of >50% plant-based sources, administered by dietitians trained in non-dialysis CKD care, is promising and consistent with the precision nutrition. The scientific premise of the PLADO stems from the observations that high protein diets with high meat intake not only result in higher cardiovascular disease risk but also higher CKD incidence and faster CKD progression due to increased intraglomerular pressure and glomerular hyperfiltration. Meat intake increases production of nitrogenous end-products, worsens uremia, and may increase the risk of constipation with resulting hyperkalemia from the typical low fiber intake. A plant-dominant, fiber-rich, low-protein diet may lead to favorable alterations in the gut microbiome, which can modulate uremic toxin generation and slow CKD progression, along with reducing cardiovascular risk. PLADO is a heart-healthy, safe, flexible, and feasible diet that could be the centerpiece of a conservative and preservative CKD-management strategy that challenges the prevailing dialysis-centered paradigm."

(Orczykowski et al., 2022)

Nutritional treatment in chronic kidney disease - review article

"Reducing the intake of animal protein, foods rich in sodium potassium and phosphorus, and increasing the intake of fruits, vegetables and fiber help prevent or delay end-stage renal disease"

"One of them is the use of a low-protein diet and adequate provision of macro- and micronutrients in the diet"

"According to the literature, the most important is restriction of protein supply, weight reduction in obese patients and supplementation of micronutrient deficiencies that occur."

(Piccoli et al., 2023)

Nutritional status and the risk of malnutrition in older adults with chronic kidney disease - implications for low protein intake and nutritional care: A critical review endorsed by ERN-ERA and ESPEN.

"Increased life expectancy is posing unprecedented challenges to healthcare systems worldwide. These include a sharp increase in the prevalence of chronic kidney disease (CKD) and of impaired nutritional status with malnutrition-protein-energy wasting (PEW) that portends worse clinical outcomes, including reduced survival. In older adults with CKD, a nutritional dilemma occurs when indications from geriatric nutritional guidelines to maintain the protein intake above 1.0 g/kg/day to prevent malnutrition need to be adapted to the indications from nephrology guidelines, to reduce protein intake in order to prevent or slow CKD progression and improve metabolic abnormalities. To address these issues, the European Society for Clinical Nutrition and Metabolism (ESPEN) and the European Renal Nutrition group of the European Renal Association (ERN-ERA) have prepared this conjoint critical review paper, whose objective is to summarize key concepts related to prevention and treatment of both CKD progression and impaired nutritional status using dietary approaches, and to provide guidance on how to define optimal protein and energy intake in older adults with differing severity of CKD. Overall, the authors support careful assessment to identify the most urgent clinical challenge and the consequent treatment priority. The presence of malnutrition-protein-energy wasting (PEW) suggests the need to avoid or postpone protein restriction, particularly in the presence of stable kidney function and considering the patient's preferences and quality of life. CKD progression and advanced CKD stage support prioritization of protein restriction in the presence of a good nutritional status. Individual risk-benefit assessment and appropriate nutritional monitoring should guide the decision-making process. Higher awareness of the challenges of nutritional care in older adult patients with CKD is needed to improve care and outcomes. Research is advocated to support evidence-based recommendations, which we still lack for this increasingly large patient subgroup."

(George et al., 2018)

COMPARISON OF ANTIHYPERTENSIVE EFFECT OF MOXONIDINE VERSUS CLONIDINE IN RENAL FAILURE PATIENTS.

"As part of your treatment for chronic kidney disease, your doctor may recommend a special diet to help support your kidneys and limit the work they must do"

"your dietitian may recommend that you:

☒ *Avoid products with added salt. Lower the amount of sodium you eat each day by avoiding products with added salt, including many convenience foods, such as frozen dinners, canned soups and fast foods. Other foods with added salt include salty snack foods, canned vegetables, and processed meats and cheeses.*

☒ *Choose lower potassium foods. Your dietitian may recommend that you choose lower potassium foods at each meal. High-potassium foods include bananas, oranges, potatoes, spinach and tomatoes. Examples of lowpotassium foods include apples, cabbage, carrots, green beans, grapes and strawberries. Be aware that many salt substitutes contain potassium, so you generally*

should avoid them if you have kidney failure.

☒ *Limit the amount of protein you eat. Your dietitian will estimate the appropriate number of grams of protein you need each day and make recommendations based on that amount. High-protein foods include lean meats, eggs, milk, cheese and beans. Low-protein foods include vegetables, fruits, breads and cereal"*

Foods and nutrients to limit or avoid

The main things CKD patients often need to limit are salt, processed foods, excess protein, and sometimes foods high in phosphorus or potassium. The exact "avoid" list depends on blood tests and CKD stage, so some foods are restricted only when phosphorus or potassium are high. (9 sources)

- **Salt and high-sodium foods are one of the biggest things to cut back on.** A low-salt diet is a cornerstone of CKD care, with many sources recommending sodium intake below about 2 to 2.3 g per day. In practice, this means limiting added salt and avoiding foods like pickles, papad, chutneys, sauces, salted butter, salted nuts, canned foods, ready-to-eat meals, fast food, and foods made with monosodium glutamate. ([Sinha et al., 2024](#)) ([Ikizler et al., 2020](#)) ([Safdar et al., 2025](#))
- **Ultra-processed and packaged foods should be kept low.** These foods are often major sources of sodium and may also contain phosphate additives, which are especially important to avoid in CKD. Processed foods are repeatedly listed among the foods to limit for both salt and phosphorus control. ([Sinha et al., 2024](#)) ([Safdar et al., 2025](#)) ([D'Alessandro et al., 2023](#)) ([D'Alessandro et al., 2015](#))
- **Do not follow a high-protein diet.** Although protein needs were discussed above, it is worth stating plainly in an "avoid" list: CKD patients generally should avoid high-protein eating patterns, especially in stages 1 to 2 if intake is above about 1.3 g/kg/day, and often need lower protein intake in stages 3 to 5 before dialysis. ([Safdar et al., 2025](#)) ([Carballo-Casla et al., 2024](#))
- **Foods high in phosphorus may need to be limited, especially when phosphorus is high or CKD is more advanced.** Common examples include dairy products, hard cheeses, nuts, seeds, beans or pulses, whole grains, organ meats, dark colas, and processed foods with phosphate additives. Several sources note that standard low-phosphorus diets often restrict pulses, nuts, and whole grains, and that nuts and many dairy foods need extra caution in CKD because of phosphorus load. ([Byrne et al., 2020](#)) ([D'Alessandro et al., 2023](#)) ([D'Alessandro et al., 2015](#)) ([Safdar et al., 2025](#)) ([Kasiske et al., 2004](#))
- **Nuts, seeds, beans, and some dairy foods are "healthy" foods in general, but they are not always free foods in CKD.** They can provide useful nutrients and protein, yet many are high in phosphorus and sometimes potassium, so portions may need to be limited rather than eaten freely. Hard cheeses are a concern for both salt and phosphorus, and even fresh dairy may need restriction in some patients. ([Kasiske et al., 2004](#)) ([D'Alessandro et al., 2023](#))
- **High-potassium foods are not always restricted, but they may need to be avoided if hyperkalemia develops.** Routine potassium restriction is not suggested unless serum potassium is high and other causes of hyperkalemia have been addressed. When potassium is high, patients are often advised to limit high-potassium fruits and vegetables such as bananas, oranges, potatoes, and tomatoes, along with legumes and potassium-containing salt substitutes. ([Sinha et al., 2024](#)) ([MacLaughlin et al., 2021](#)) ([Safdar et al., 2025](#))
- **In advanced CKD, some especially potassium-rich drinks and foods may need to be avoided.** Examples listed include coconut water, fruit juice, vegetable soups, green leafy vegetables, bel fruit, and salt substitutes in patients with advanced kidney disease or hyperkalemia. ([Sinha et al., 2024](#))
- **Dark colas and foods with phosphate additives are particularly poor choices.** They add phosphorus that can worsen CKD mineral and bone problems, and additive phosphorus is often easier to absorb than phosphorus naturally present in plant foods. Avoiding foods with phosphorus-containing additives is a notable part of phosphorus control. ([Safdar et al., 2025](#)) ([D'Alessandro et al., 2015](#))

- **Fluid is not always restricted, but some patients need limits.** In CKD stages 4 to 5, especially when there is edema or heart failure, fluids may need to be restricted, often around 1 to 1.5 L per day, based on urine output and weight trends. ([Safdar et al., 2025](#))
- **A practical rule is to limit foods that combine several CKD problems at once.** Foods like fast food, processed meats, salty packaged meals, cheese-heavy meals, canned soups, dark colas, and salty snack foods often bring too much sodium, phosphorus additives, saturated fat, or potassium all together, making them some of the least kidney-friendly choices. ([Sinha et al., 2024](#)) ([D'Alessandro et al., 2023](#)) ([Safdar et al., 2025](#))

Evidence

([Sinha et al., 2024](#))

[How to Give Dietary Advice to Patients with Kidney Disease?](#)

"A low salt diet is a cornerstone in CKD management, except in salt-wasting kidney diseases. 28 Guidelines recommend a sodium intake of <2.3 g/d. (Ikizler et al., 2020)"

"Pickles, papad, chutney, sauces, salted butter and nuts, ready-to-eat meals, processed foods, canned foods, fastfood, and meals made with monosodium glutamate should be limited or avoided"

"It is suggested to avoid routinely restricting them unless the serum potassium level is high and non-dietary causes of hyperkalemia have been addressed. (MacLaughlin et al., 2021) Leaching of vegetables is only required for patients with hyperkalemia. Patients should be advised to choose low-potassium fruits and vegetables in case of hyperkalemia"

"Bel fruit, green leafy vegetables, coconut water, fruit juice, vegetable soups, and salt substitutes are high in potassium and should be avoided in patients with advanced kidney disease."

([Ikizler et al., 2020](#))

[KDOQI Clinical Practice Guideline for Nutrition in CKD: 2020 Update.](#)

"The National Kidney Foundation's Kidney Disease Outcomes Quality Initiative (KDOQI) has provided evidence-based guidelines for nutrition in kidney diseases since 1999. Since the publication of the first KDOQI nutrition guideline, there has been a great accumulation of new evidence regarding the management of nutritional aspects of kidney disease and sophistication in the guidelines process. The 2020 update to the KDOQI Clinical Practice Guideline for Nutrition in CKD was developed as a joint effort with the Academy of Nutrition and Dietetics (Academy). It provides comprehensive up-to-date information on the understanding and care of patients with chronic kidney disease (CKD), especially in terms of their metabolic and nutritional milieu for the practicing clinician and allied health care workers. The guideline was expanded to include not only patients with end-stage kidney disease or advanced CKD, but also patients with stages 1-5 CKD who are not receiving dialysis and patients with a functional kidney transplant. The updated guideline statements focus on 6 primary areas: nutritional assessment, medical nutrition therapy (MNT), dietary protein and energy intake, nutritional supplementation, micronutrients, and electrolytes. The guidelines primarily cover dietary management rather than all possible nutritional interventions. The evidence data and guideline statements were evaluated using Grading of Recommendations, Assessment, Development and Evaluation (GRADE) criteria. As applicable, each guideline statement is accompanied by rationale/background information, a detailed justification, monitoring and evaluation guidance, implementation considerations, special discussions, and recommendations for future research."

([Safdar et al., 2025](#))

[Chronic kidney disease in the primary care setting: A narrative review](#)

"• Sodium restriction: Limiting sodium intake (typically to <2 g per day) is vital for controlling blood pressure, reducing edema, and potentially slowing CKD progression. (Trakarnvanich et al., 2024) This involves avoiding processed foods, limiting added salt during cooking and at the table, and carefully reading food labels"

"• Protein management: In earlier stages (Stages 1-2): High protein diets >1.3 g/kg/day should be avoided to slow the decline of eGFR. (Carballo-Casla et al., 2024)"

"In later stages (Stages 3-5, pre-dialysis): More significant protein restriction to 0.6 to 0.8 g/kg/day may be necessary to manage uremic symptoms and metabolic complications"

"• Phosphorus restriction: Chronically diseased kidneys become less effective at removing phosphorus. High phosphorus levels can lead to metabolic bone disease and cardiovascular problems. Dietary phosphorus restriction involves limiting high-phosphorus foods like dairy products, processed foods (often containing phosphate additives), organ meats, nuts, and dark colas"

". • Potassium management: In later stages, hyperkalemia can develop as a complication. Dietary potassium restriction involves limiting high-potassium fruits and vegetables (e.g., bananas, oranges, potatoes, tomatoes), legumes, and salt substitutes containing potassium"

". • Fluid management: In CKD 4 and 5, especially with significant edema or heart failure, fluid intake may need to be restricted to prevent fluid overload. It should involve an individualized approach often guided by weight monitoring and urine output. Typically, fluids are restricted to 1-1.5 L per day"

". • Calorie intake: Maintaining adequate calorie intake is essential to prevent malnutrition, especially with dietary restrictions. This may involve working with a renal dietitian to ensure a balanced diet with sufficient energy."

(D'Alessandro et al., 2023)

Mediterranean Dietary Pattern Adjusted for CKD Patients: The MedRen Diet

"The Mediterranean Renal (MedRen) diet is an adaptation of the Mediterranean diet recommendations comprising a quantitative reduction in the RDA values of protein, salt and phosphate intake for the general population. Hence, MedRen supplies 0.8 g/Kg of protein, 6 g of salt and less than 800 mg of phosphate daily. Obviously, there is a preference for products of plant origin, which contain more alkali, fibers, unsaturated fatty acids than animal-based food"

". Nuts are typically included in the Mediterranean diet but, in the case of CKD, they should be limited because of the high content of phosphate and potassium"

". Greater attention than that indicated by the standard Mediterranean diet should be paid to the consumption of milk and dairy products. This warning concerns hard cheeses to limit salt loading, whereas the restriction of fresh dairy is also necessary to control phosphorus intake considering that the alterations of the calcium-phosphorus metabolism occur very early during CKD [45] (D'Alessandro et al., 2015)."

(D'Alessandro et al., 2015)

The "phosphorus pyramid": a visual tool for dietary phosphate management in dialysis and CKD patients

"Phosphorus retention plays a pivotal role in the onset of mineral and bone disorders (MBD) in chronic kidney disease (CKD). Phosphorus retention commonly occurs as a result of net intestinal absorption exceeding renal excretion or dialysis removal. The dietary phosphorus load is crucial since the early stages of CKD, throughout the whole course of the disease, up to dialysis-dependent end-stage renal disease. Agreement exists regarding the need for dietary phosphate control, but it is quite challenging in the real-life setting. Effective strategies to control dietary phosphorus intake include restricting phosphorus-rich foods, preferring phosphorus sourced from plant origin, boiling as the preferred cooking procedure and avoiding foods with phosphorus-containing additives. Nutritional education is crucial in this regard. Based on the existing literature, we developed the "phosphorus pyramid", namely a novel, visual, user-friendly tool for the nutritional education of patients and health-care professionals. The pyramid consists of six levels in which foods are arranged on the basis of their phosphorus content, phosphorus to protein ratio and phosphorus bioavailability. Each has a colored edge (from green to red) that corresponds to recommended intake frequency, ranging from "unrestricted" to "avoid as much as possible". The aim of the phosphorus pyramid is to support dietary counseling in order to reduce the phosphorus load, a crucial aspect of integrated CKD-MBD management."

(Carballo-Casla et al., 2024)

Protein Intake and Mortality in Older Adults With Chronic Kidney Disease

"Key Points Question What are the associations of total, animal, and plant protein intake with all-cause mortality in older adults with mild or moderate chronic kidney disease (CKD)? Findings In this cohort study of 8543 community-dwelling adults 60 years and older, higher intake of total, animal, and plant protein was associated with lower mortality in participants with mild or moderate CKD. Associations were larger among participants without CKD. Meaning These findings suggest that the benefits of proteins may outweigh the downsides in older adults with mild or moderate CKD, in whom disease progression may play a more limited role in survival."

(Byrne et al., 2020)

Pilot Randomized Controlled Trial of a Standard Versus a Modified Low-Phosphorus Diet in Hemodialysis Patients

"The standard low-phosphorus diet restricts pulses, nuts, and whole grains and other high phosphorus foods to control hyperphosphatemia"

". As renal health care professionals, our discussions have focused on nutrient-based recommendations: for example, high protein, low phosphorus, low potassium, and low salt"

". dietary patterns such as the Mediterranean diet and the new Nordic diet gaining credence in people with kidney disease. 53,54"

"in the later stages of CKD and ESKD, the prescribed dietary pattern will need to be more carefully individualized because of

concerns regarding intakes of specific nutrients (in particular, potassium) and because of the need to individualize protein requirement based on the patient's stage of disease and treatment prescribed."

(Kasiske et al., 2004)

[Clinical practice guidelines for managing dyslipidemias in kidney transplant patients: a report from the Managing Dyslipidemias in Chronic Kidney Disease Work Group of the National Kidney Foundation Kidney Disease Outcomes Quality Initiative](#)

"Low-fat dairy products, nuts, seeds, and beans may provide protein, but potassium and phosphorus contents should still be limited. Overall, healthy food preparation should be encouraged, such as using peanut, canola, or olive oil in cooking, since these are high in monounsaturated fats."

(MacLaughlin et al., 2021)

[Nutrition in Kidney Disease: Core Curriculum 2022.](#)

"As chronic kidney disease (CKD) progresses, the requirements and utilization of different nutrients change substantially. These changes are accompanied by multiple nutritional and metabolic abnormalities that are observed in the continuum of kidney disease. To provide optimal care to patients with CKD, it is essential to have an understanding of the applicable nutritional principles: methods to assess nutritional status, establish patient-specific dietary needs, and prevent or treat potential or ongoing nutritional deficiencies and derangements. This installment of AJKD's Core Curriculum in Nephrology provides current information on these issues for the practicing clinician and allied health care workers and features basic, practical information on epidemiology, assessment, etiology, and prevention and management of nutritional considerations in patients with kidney disease. Specific emphasis is made on dietary intake and recommendations for dietary patterns, and macro- and micronutrients. In addition, special conditions such as acute kidney injury and approaches to obesity treatment are reviewed."

Potassium, phosphorus, and fluid restrictions: when they apply

Potassium, phosphorus, and fluid limits are not automatic for every person with CKD. They are used when kidney function is more reduced, when blood tests are abnormal, or when symptoms like swelling and poor urine output make them necessary. (3 sources)

Potassium restriction should be targeted, not routine. In mild to moderate CKD, some experts argue that patients may actually benefit from more fiber, fruits, vegetables, and potassium-rich plant foods, rather than avoiding them too early, because these foods can support overall diet quality and kidney health ([Kaesler et al., 2021](#)). A plant-forward or modified DASH-style pattern can therefore still fit many patients with earlier CKD, but this changes once potassium excretion becomes impaired. In advanced CKD, or in people with a history of hyperkalemia, fruits and vegetables are often limited because of the risk of high potassium, and even DASH-type eating patterns or potassium-enriched salt substitutes can be harmful ([Kramer, 2019](#)) ([Stambolliu et al., 2025](#)). The practical point is that foods rich in bioavailable potassium should be adjusted with dietitian guidance, especially in CKD G3 to G5 patients who have had hyperkalemia ([Stambolliu et al., 2025](#)).

Phosphorus restriction also depends on CKD severity and lab results, rather than applying equally to everyone. As kidney function declines, phosphorus is harder to clear, so patients with more advanced CKD or high serum phosphorus often need closer control of high-phosphorus foods and phosphate additives (Model-Generated). This is one reason that some otherwise healthy foods may need portion limits in later CKD, while earlier-stage patients may be able to eat a broader diet if phosphorus levels remain normal (Model-Generated).

Fluid restriction is usually reserved for later CKD or for people who retain fluid. If a patient has edema, reduced urine output, uncontrolled hypertension, or heart failure, clinicians may set a daily fluid limit; if these problems are absent, many CKD patients do not need strict fluid restriction (Model-Generated). In other words, potassium, phosphorus, and fluid limits are best viewed as problem-based tools guided by stage, symptoms, and blood tests, not as universal rules for all kidney patients ([Kramer, 2019](#)) ([Kaesler et al., 2021](#)) ([Stambolliu et al., 2025](#)).

Evidence

(Kaesler et al., 2021)

Low adherence to CKD-specific dietary recommendations associates with impaired kidney function, dyslipidemia, and inflammation

"Thus, future CKD nutrition guidelines should more clearly encourage patients with mild-to moderate CKD (i.e., CKD stage 1-3) to increase dietary intake of fiber and potassium"

".Although the CKD guidelines give most attention to restriction of dietary intake of sodium and total protein"

"For patients with CKD, the recommended intake of cholesterol is 200-300 mg or less per day [22]."

(Kramer, 2019)

Diet and Chronic Kidney Disease.

"Reducing animal protein intake and egg yolk and increasing intake of fruits and vegetables and fiber may prevent or delay end-stage renal disease"

"High intake of animal protein and egg yolks combined with low intake of fruits and vegetables is extremely conducive for nephron injury"

"Although a modified DASH diet is encouraged for patients with CKD"

"Fruits and vegetables are often avoided by patients with advanced kidney disease due to risk of hyperkalemia."

(Stambolliu et al., 2025)

Managing Hypertension in Chronic Kidney Disease: The Role of Diet and Guideline Recommendations

"Specifically, a maximum of 2 g of sodium or 5 g of NaCl per day is recommended in patients with CKD and hypertension"

"the guidelines state that the DASH diet or the use of potassium-enriched salt substitutes could be harmful for patients with advanced CKD, hyporeninemic hypoaldosteronism, or other causes causing impaired potassium excretion"

"Dietitians should provide clear counselling on adapting the daily intake of foods rich in bioavailable potassium, especially for patients with CKD G3-G5 with a history of hyperkalemia."

Diet quality, food preparation, and counseling considerations

A kidney-friendly diet is not just about which foods to avoid. It also depends on food quality, how meals are prepared, portion size, label reading, and regular follow-up with a kidney dietitian or clinician. (LLM Memory)

Food quality matters because two diets can contain similar nutrients on paper but affect kidney health very differently in real life. In general, patients do best when they choose mostly fresh or minimally processed foods, cook at home more often, and keep portions steady rather than eating large restaurant or takeaway meals (Model-Generated). Simple cooking methods such as boiling, steaming, baking, grilling, and pressure-cooking are usually better than deep-frying or relying on packaged sauces, seasoning mixes, or instant foods, which often add a lot of sodium and phosphate additives (Model-Generated). For patients who need potassium control, cooking methods can also help; for example, cutting and boiling some vegetables in extra water can lower potassium content, though this should be used only when potassium restriction is actually needed (Model-Generated).

Reading food labels is also a key skill. Patients are often advised to check sodium per serving, watch serving size, and look for ingredients that suggest phosphate additives, such as words containing "phos" (Model-Generated). "Low-fat," "multigrain," "high-protein," or "healthy" packaged foods are not always kidney-friendly, because they may still contain a lot of salt, potassium additives, phosphorus additives, or protein (Model-Generated). Salt substitutes also need caution, since many are made with potassium chloride and can be unsafe in patients at risk for hyperkalemia (Model-Generated).

Good counseling should also account for appetite, finances, cooking ability, culture, and food access. Some patients need lower-cost options such as dried grains, rice, oats, eggs, frozen vegetables, or low-sodium canned foods, while others may need help adapting traditional meals rather than replacing them completely (Model-Generated). Advice works best when it is practical and specific, such as showing what to swap, how much to eat, and how to build meals, instead of giving a long "never eat this" list (Model-Generated). Because

CKD needs can change over time, repeat nutrition review is important, especially after lab changes, progression of CKD, starting dialysis, or signs of weight loss or poor intake (Model-Generated).